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ABSTRACT

Enhancing the horizontal mobility of coalesced droplets on a plane could promote droplet jumping. Here, we achieve enhanced horizontal mobility of a coalesced jumping droplet on superhydrophobic surfaces with an asymmetric ridge and investigate the underlying mechanism through experiment and simulation. Results indicate that the coalesced droplet accelerates during the coalescence-induced jumping stage and gains horizontal velocity during the rebound stage. The nondimensional horizontal velocity can reach 0.47, which is about 2.3 times the jumping velocity on the plane. Depending on the height-to-width ratio of the asymmetric ridge, the ratio of the horizontal velocity to the fallen velocity when the fallen droplet makes contact with the ridge is 0.55–0.75. Furthermore, the coalesced droplet can still obtain considerable horizontal velocity on superhydrophobic surfaces with an asymmetric ridge when the initial droplet radius is unequal. This work provides new insights for improving droplet jumping by surface structure in related fields.

I. INTRODUCTION

Spontaneous jumping triggered by the coalescence of two or more adjacent droplets on superhydrophobic surfaces (SHSs) can significantly promote the self-removal of droplets,[1] which has broad potential application in the fields of enhanced heat transfer,[2–4] self-cleaning,[5] energy harvesting,[6,7] and anti-icing/frosting.[8–10] Many studies have investigated droplet jumping from the perspectives of jumping velocity and direction,[11,12] energy conversion,[13–17] and size effects.[18–22] For the coalescence of two equal-size droplets on the plane, the jumping velocity V_j follows the inertial-capillary scaling:[23] $V_j = \varphi V_{ci}$, where $V_{ci} = (\gamma_{lv}/\rho_l R)^{0.5}$ is the inertia-capillary velocity, γ_{lv} is the surface tension, ρ_l is the fluid density, R is the initial droplet radius, and φ is a constant between 0.2 and 0.3.[1,9] Energy analyses show that the surface energy released during coalescence is partially converted into jumping kinetic energy with a low conversion rate η of only about 6%.[24–26] For the coalescence of unequal-size droplets, the jumping velocity, released surface energy, and energy conversion rate will decrease and jumping will not occur when the droplet radius ratio is less than a critical value.[20,21,27] Yan *et al.*[27] found through experiments that the jumping direction of the coalesced droplet is perpendicular to the plane, which has no apparent relationship with the droplet radius ratio. For the multi-droplet coalescence of three equal-size droplets, the nondimensional jumping velocity is enhanced,[28] but the energy conversion rate is still in the low range (below 10%) with different initial arrangements of droplets, droplet numbers, and initial radius ratios.[13,14,29,30] Moreover, the jumping direction of multi-droplet jumping remains perpendicular to the plane.[13] These results show that the energy conversion rate of droplet jumping on the plane is low and that the jumping direction is perpendicular to the plane, which limits droplet jumping performance in practical applications.

The jumping velocity and energy conversion rate can be greatly enhanced by applying macrostructures. Wang *et al.*[31] achieved an enhanced nondimensional jumping velocity of 0.53 with an energy conversion rate of 22.5% using a triangular prism. Vahabi *et al.*[32] demonstrated that the enhanced energy conversion rate results from the redirection of the internal flow. Yuan *et al.*[33] proposed an "egg-tray" structure that could achieve the theoretical maximum jumping velocity ($V_j \approx 0.78 V_{ci}$) and energy conversion rate ($\eta \approx 52.4\%$) through an artificial neural network. Furthermore, macrostructures can realize both enhancement of the jumping velocity and control of the jumping direction. Peng *et al.*[34] achieved a maximum nondimensional jumping velocity of about 0.74 with an energy conversion rate of about 46% using micro-grooves, and the jumping direction deviated from the plane normal by 35°. The deviation in the jumping direction is caused by a change in the direction of the reaction force between the liquid bridge and the surface.[35] Liu *et al.*[36] further increased the deviation in the jumping direction from the surface normal to 73° with enhanced energy conversion of 46% using a U-groove structure. In addition, curved surfaces,[37] asymmetric V-grooves,[38] strings,[39] nano-grassed zig-zags,[40] micro-grooves,[41] and V-shaped grooves[42] have the ability to break the jumping velocity limit and regulate the jumping direction. The deviation in the jumping direction gives the jumping droplet horizontal velocity, which can increase the coalescence frequency between surrounding droplets. In practical applications, frequent droplet jumping caused by droplet coalescences can promote the self-removal of droplets and significantly improve performance.

On the plane, the jumping droplet falls back to the surface under the force of gravity,[43] which means it makes contact with the surface twice: first at the time of coalescence and then while falling back. Previous studies on the regulation of droplet jumping using macrostructures have mainly focused on the droplet coalescence stage; the effects of macrostructures on the motion of a coalesced jumping droplet that falls back to the surface have been investigated less. The process of a droplet falling back to the surface is similar to that of a fallen droplet that impacts the surface at a low *We*, which has been extensively studied. Macrostructures such as ridges,[44–50] point-like defects,[51] tapered post arrays,[52] ring surfaces,[53] cone,[54] craterlike surface,[55] and concave surfaces[56] have notable effects on the dynamic process of droplet impact. However, most studies focus on the contact time, spreading length, and morphological variation in the impacting droplet; investigations of the horizontal motion of the impacting droplet are lacking. In terms of droplet motion, a horizontal moving surface can make the impacting droplet have a horizontal displacement.[57] A droplet deposited on the surface can be removed by the impact from another droplet, but horizontal motion is limited.[58–61] Moreover, adjusting the wettability can result in the transport of the impacting droplet, and the transport distance can reach up to tens of times the droplet radius.[62–64] These studies only focus on impacting droplets and do not consider the effect of macrostructures on the droplet motion. Therefore, it is necessary to investigate the effect of macrostructures on the horizontal motion of the jumping droplet that falls back to the surface.

In this work, we fabricate superhydrophobic surfaces decorated with an asymmetric ridge (AR) and report that the horizontal mobility of a droplet can be enhanced when the coalesced jumping droplet falls back to the surface and rebounds under the influence of the asymmetric ridge. The droplet motion on the plane is observed experimentally and is divided into three stages: coalescence-induced jumping, detachment, and rebound. The evolution of the droplet morphology, the velocity, and the effect of geometric parameters of the asymmetric ridge are studied experimentally. Numerical simulations are performed to investigate the enhancement mechanism, droplet dynamics, and energy conversion during the coalescence-induced jumping and rebound stages. Furthermore, we demonstrate experimentally that the enhancement of horizontal mobility is effective for a coalesced droplet with unequal initial radii.

II. METHODS

A. Fabrication of superhydrophobic surfaces with an asymmetric ridge

Figure 1(a) shows the schematic of the asymmetric ridge. The ridge is fabricated on aluminum substrate by a computer numerical control (CNC) milling machine, as shown in Fig. 1(b). Based on the minimum processing size of the CNC mill and the droplet size in experiments, the bottom width W of the asymmetric ridge used in the experiments is 0.2 mm, and the heights H are 0.2, 0.3, and 0.4 mm, respectively. We form the superhydrophobic surfaces by spraying superhydrophobic nano-coating (Schanda, China) in a two-step process. First, the substrates are put into de-ionized water and ethanol for

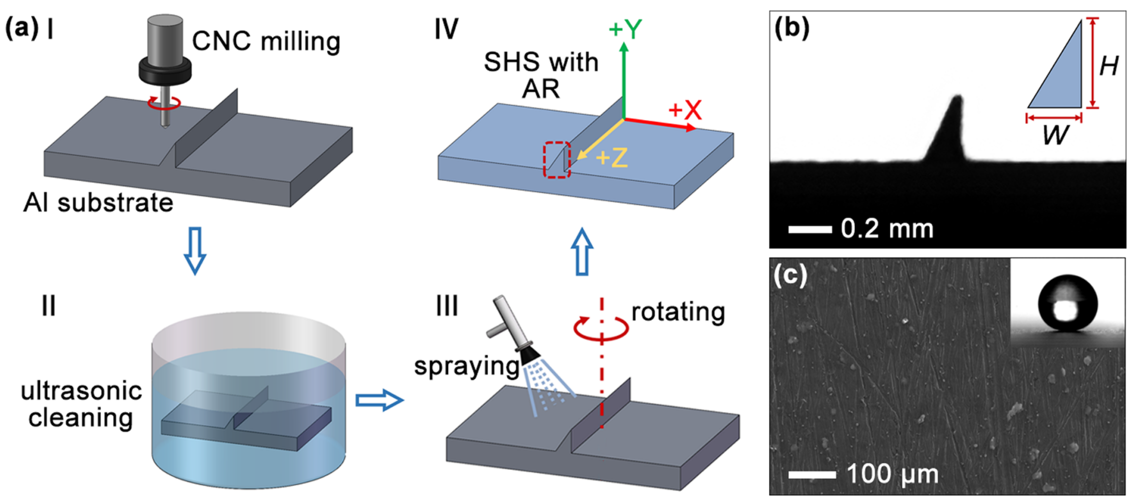

FIG. 1. (a) Fabrication of a superhydrophobic surface (SHS) with an asymmetric ridge (AR). The substrate is aluminum (Al). (b) Side view of the AR. H refers to the height of the ridge, and W refers to the width. (c) Scanning Electron Microscope (SEM) image of the SHS. The contact of a 2 μl water droplet shows the good superhydrophobicity of the surface. SEM, CNC, Al.

ultrasonic cleaning for 20 min and then dried under the ambient condition (ambient temperature $T_a = 25 \pm 2\,^\circ$C, relative humidity $RH = 50\% \pm 10\%$). Second, the superhydrophobic nano-coating is sprayed on the cleaned substrates uniformly from a distance of about 30 cm, and the airflow direction is about 45° from the plane. During spraying, the substrate is rotated 45° every 5 s, and this is repeated 16 times. After spraying, the substrate is dried at room temperature for 24 h. The static contact angle and rolling angle of a 2 μl water droplet are 160° ± 1° and 2° ± 1°, respectively (JC2000C1, China). An SEM image of a superhydrophobic surface is shown in Fig. 1(c).

B. Experimental setup and numerical simulation

A high-speed visualization platform is built to investigate the motion of the coalesced jumping droplet. The platform includes a triple-axis stage, an LED light source, a microinjector, and a high-speed camera, as shown in Fig. 2(a). The droplet coalescence process is recorded by the high-speed camera (Revealer X113M; Fuhuang Agile Device) equipped with a macrolens (LAOWA 60 mm F2.8 2:1) at a frame rate of 5000 fps. Given the maximum volume (5 μl) of the microinjector and the droplet size in the previous studies,[35,36,38] the initial droplet radius in the experiments ranges from 0.66 to 1.25 mm, with a corresponding volume of 1.2 to 8.2 μl. At the beginning of the experiment, the substrate is fixed on the triple-axis stage, and the first droplet generated by the microinjector is placed near the ridge. The triple-axis stage and light source are adjusted to get a clear view of the droplet on the camera. Then, the second droplet is placed on the substrate using the same method and is gently pushed by a molybdenum needle sprayed with the same superhydrophobic coating to induce droplet coalescence. The trajectory of the coalesced droplet is obtained by the ImageJ software, and all the velocities are calculated from the trajectory. The accuracy varies from 0.012 68 to 0.015 07 mm depending on each experiment. All experiments are repeated six times under the ambient condition.

A user-defined jumpingFoam solver is used to simulate coalescence-induced jumping and rebound in the droplet motion. Based on the interFoam solver in the OpenFOAM, the jumpingFoam solver uses the volume of the fluid method to track the phase interface and the continuum surface force model to calculate surface tension. In addition, functions about the calculation of energy terms such as total kinetic energy, jumping kinetic energy, and viscous dissipation are added to the modified solver. The jumpingFoam solver has been used and validated in our previous studies, and more details are also included.[33,39] The computational domain has a size of $6.4 \times 6 \times 2.4\,\text{mm}^3$ with 78×10^6 polyhedral meshes, and the geometric parameter of the ridge is $W = 0.2$ mm, $H = 0.4$ mm. The symmetry boundary condition is set to conserve computational resources, with a pressure-outlet boundary for the top boundary condition and wall boundary with a static contact angle of 160° for the other boundary conditions, as shown in Fig. 2(b). For the coalescence-induced jumping stage, the initial droplet radius R is 0.8 mm. For the rebound stage, the droplet volume is equal to the volume of the two initial droplets with an equivalent radius $R_e = 2^{1/3}R = 1.007$ mm. The experimental results show that the initial droplet has a horizontal offset Δx [Fig. 3(a)], and the relative horizontal offset $\Delta x/R$ in the simulation is 0.4625, which is the same as that of $R = 0.8$ mm. In practice, the droplet shape is oscillatory and is not an ideal sphere when the droplet falls back to the surface. However, the jumping process is a low deformation process, as We is less than 1, so, the fallen droplet is treated as an ideal sphere.[45] The two stages are simulated using the same computational domain and physical parameters. The density of water ρ_l is 997 kg/m^3, the viscosity of water μ_l is 8.937×10^{-4} Pa s, the density of air ρ_v is 1 kg/m^3, the viscosity of air μ_v is 1.834×10^{-5}, and the surface tension γ_{lv} is 0.072 N/m.

The vertical and horizontal components V_y and V_x of jumping velocity $\boldsymbol{V}$ and the vertical and horizontal components F_y and F_x of reaction force $\boldsymbol{F}$ are calculated as follows:[37,65]

$$V_y = \frac{\int_\Omega \rho_l v \, d\Omega}{\int_\Omega \rho_l \, d\Omega}, \tag{1}$$

$$V_x = \frac{\int_\Omega \rho_l u \, d\Omega}{\int_\Omega \rho_l \, d\Omega}, \tag{2}$$

$$F_y = \frac{d\left(\int_\Omega \rho_l v \, d\Omega\right)}{dt}, \tag{3}$$

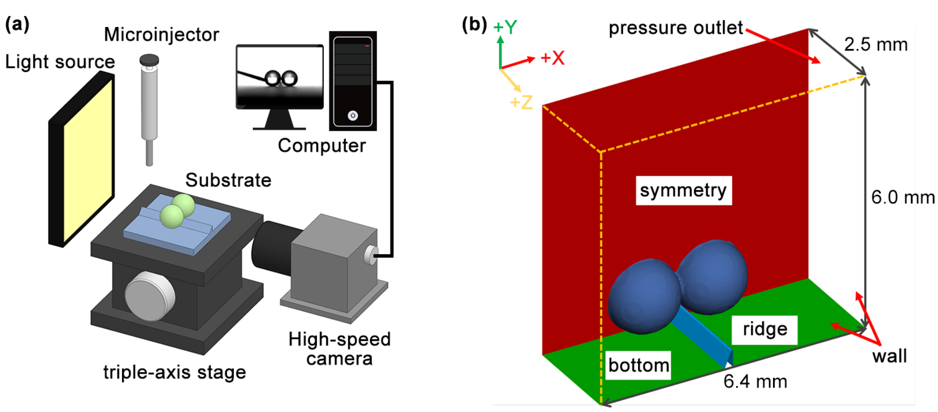

FIG. 2. (a) Schematic of the experimental platform. (b) Schematic of the computational domain and boundary conditions.

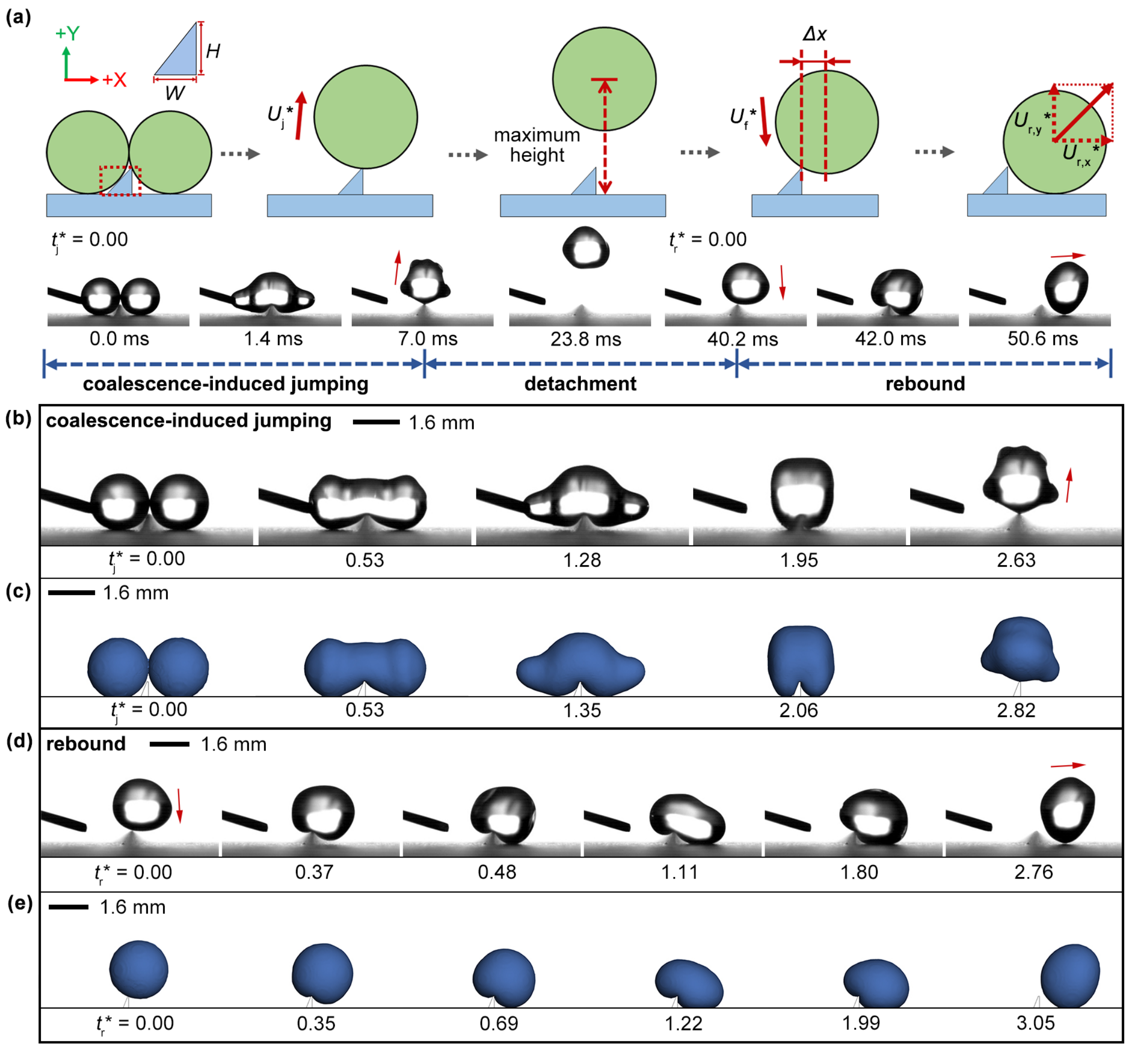

FIG. 3. (a) Schematic of droplet motion and its three stages. Snapshots of droplet morphology in the coalescence-induced jumping stage: (b) experimental results and (c) numerical results. Snapshots of droplet morphology in the rebound stage: (d) experimental results and (e) numerical results. Multimedia view: https://doi.org/10.1063/5.0121402.1

$$F_x = \frac{\mathrm{d}\left(\int\limits_{\Omega} \rho_l u \mathrm{d}\Omega\right)}{\mathrm{d}t}, \tag{4}$$

where Ω is the domain of the droplet and u, v, and w are the velocity components of the internal flow in the X, Y, and Z directions, respectively. The vertical and horizontal components of the jumping velocity, reaction force, and time are nondimensionalized as follows:[66,67]

$$U_y^* = \frac{V_y}{\sqrt{\gamma_{lv}/(\rho_l R)}}, \tag{5}$$

$$U_x^* = \frac{V_x}{\sqrt{\gamma_{lv}/(\rho_l R)}}, \tag{6}$$

$$F_y^* = \frac{F_y}{4\rho_l V_m{}^2 R}, \tag{7}$$

$$F_x^* = \frac{F_x}{4\rho_l V_m{}^2 R}, \tag{8}$$

$$t^* = \frac{t}{\sqrt{\rho_l R^3/\gamma_{lv}}}, \tag{9}$$

where V_m is the jumping velocity in the coalescence-induced jumping stage, and the fallen velocity in the rebound stage.

III. RESULTS AND DISCUSSION

A. Motion of the coalesced jumping droplet

Figure 3(a) (Multimedia view) shows the motion of the coalesced jumping droplet when the height-to-width ratio H/W is 2.0. The entire process can be divided into three stages: coalescence-induced jumping, detachment, and rebound. The coalescence-induced jumping stage is when the adjacent droplets coalesce and move up to the top of the ridge. The detachment stage is when the coalesced jumping droplet moves upward from the ridge to reach the maximum height, falls back, and then makes contact with the ridge again under the force of gravity. The rebound stage is when the fallen droplet impacts the ridge eccentrically and rebounds. Because the droplet does not make contact with the surface during the detachment stage, this work focuses on the coalescence-induced jumping and rebound stages. Figures 3(b)–3(e) show experimental results ($R = 0.8$ mm, $H/W = 2.0$) and numerical results for the two stages of motion of the droplet. Because of the long duration of the droplet motion, the two stages when the droplet makes contact with the surface are simulated. The time when the fallen droplet impacts the ridge is taken as the initial moment of the simulation in the rebound stage. The time is nondimensionalized to compare the experimental and numerical results, where t^* is the nondimensional time of the experimental process and t_j^* and t_r^* are the nondimensional times of the coalescence-induced jumping stage and the rebound stage, respectively. Figures 3(b) and 3(c) show the experimental and simulation results for the morphological evolution of droplets in the coalescence-induced jumping stage. At $t_j^* = 0$, adjacent droplets make contact with each other and begin to coalesce. Then at $t_j^* = 0.53$, the expanding liquid bridge makes contact with the ridge, resulting in symmetry breaking in the expansion of the liquid bridge. At $t_j^* = 1.28$, the liquid on both sides of the coalescing droplet contracts toward the middle. At $t_j^* = 1.95$, the coalescing droplet wraps around the ridge. The interaction force generated by the impact of the liquid bridge and the surface causes the coalescing droplet to move upward and eventually detach from the surface ($t_j^* = 2.63$). Figures 3(d) and 3(e) show the experimental and simulation results for the morphological evolution of the droplet in the rebound stage. At $t_r^* = 0$, the fallen droplet impacts the top of the ridge eccentrically. Then, the droplet continues to move downward but does not make contact with the plane, with the bottom of the droplet deformed because of the extrusion of the ridge ($t_r^* = 0.37$). At $t_r^* = 0.48$, the droplet makes contact with the plane, and the droplet morphology shows a clear asymmetry (i.e., the left side is higher than the right side) due to the eccentric impact of the fallen droplet and the ridge. At $t_r^* = 1.11$, the droplet shows obvious asymmetric spreading under the effect of the ridge, which leads to asymmetric contraction of the spreading droplet ($t_r^* = 1.80$). At $t_r^* = 2.76$, the droplet detaches from the ridge and has the significant horizontal velocity.

Figure 4 shows the experimental results for velocity in droplet motion with different initial droplet radii on superhydrophobic surfaces with asymmetric ridges of varying height-to-width ratios.

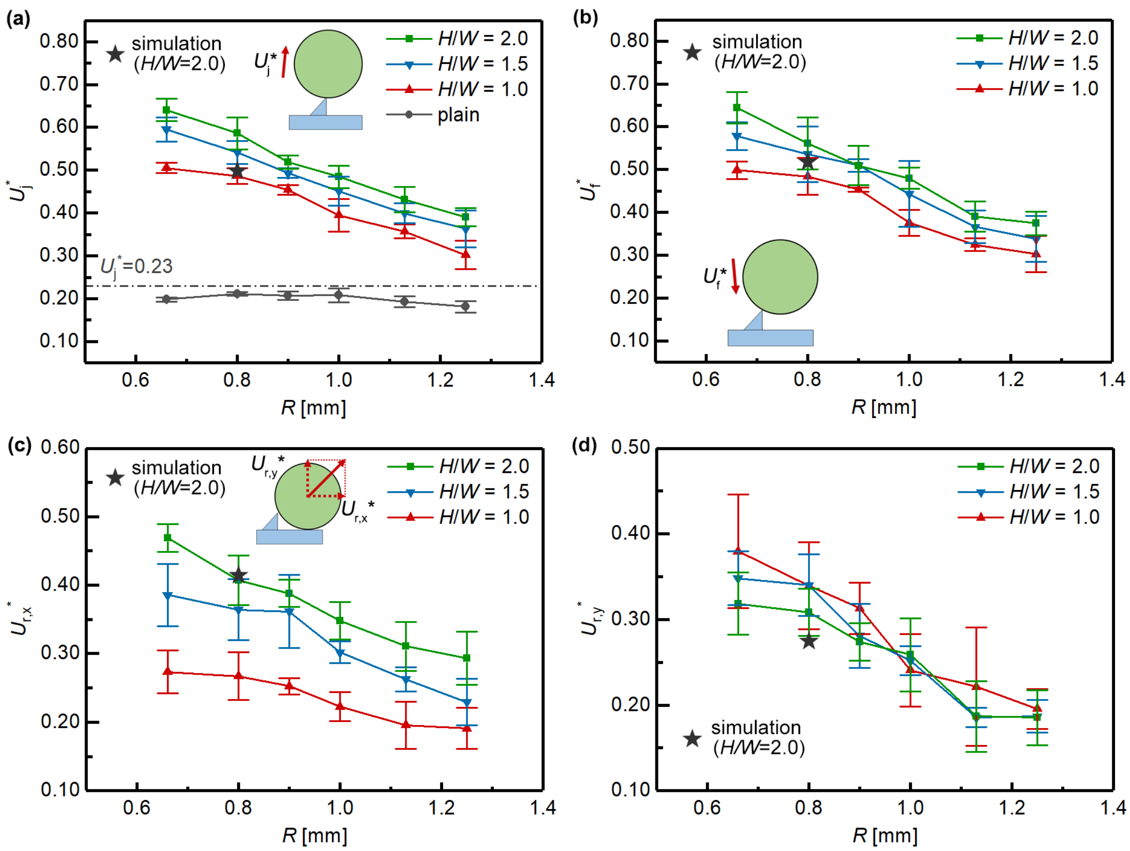

FIG. 4. Variation in the experimentally obtained droplet velocity in droplet motion with an initial droplet radius R and height-to-width ratio H/W: (a) nondimensional jumping velocity U_j^*, (b) nondimensional fallen velocity U_f^*, (c) nondimensional horizontal velocity $U_{r,x}^*$, and (d) nondimensional vertical rebound velocity $U_{r,y}^*$. The star denotes the numerical results.

The initial droplet radius R ranges from 0.66 to 1.25 mm, and the geometric parameters of the asymmetric ridge are $W = 0.2$ mm and $H/W = 1.0$, 1.5, and 2.0. The nondimensional jumping velocity U_j^* is the nondimensional velocity when the coalesced jumping droplet is about to detach from the top of the ridge. As shown in Fig. 4(a), U_j^* increases with the decrease in R and the increase in H/W. During the coalescence on the superhydrophobic surfaces with the ridge, the velocity vectors of the internal flow are redirected upward, resulting in enhanced jumping velocity.[32] As R decreases or H/W increases, the ratio of ridge height to droplet radius H/R increases. According to Ref. 33, the additional surface energy increases with the increase of H/R, which leads to larger jumping velocity. U_j^* reaches the maximum value of 0.64 when $R = 0.66$ mm and $H/W = 2.0$ and the minimum value of 0.31 when $R = 1.25$ mm and $H/W = 1.0$. Compared to the nondimensional jumping velocity ($U_j^* \approx 0.20$) of the jumping on the plane, the jumping velocity is increased by 50% to 235% depending on the different initial droplet radii. Figure 4(b) shows the variation in nondimensional fallen velocity U_f^* with R and H/W when the fallen droplet impacts the ridge. The nondimensional falling velocity U_f^* is the nondimensional velocity when the fallen jumping droplet is about to contact the top of the ridge. U_f^* shows an increasing trend with the decrease in R or the increase in H/W, which is similar to the change in U_j^*. Because the height of the droplet is nearly the same when the jumping droplet detaches from the surface and when the fallen droplet impacts the ridge, the change in the gravitational potential energy can be ignored, which indicates that U_f^* and U_j^* are basically the same. To illustrate the ability of the horizontal motion, Figs. 4(c) and 4(d) show variation in the nondimensional horizontal velocity $U_{r,x}^*$ and the nondimensional vertical rebound velocity $U_{r,y}^*$ with R and H/W when the fallen droplet rebounds from the ridge. The nondimensional horizontal velocity $U_{r,x}^*$ is the nondimensional velocity in the X direction when the fallen droplet is about to rebound from the ridge, and the nondimensional vertical rebound velocity $U_{r,y}^*$ is the nondimensional velocity in the Y direction when the fallen droplet is about to rebound from the ridge. $U_{r,x}^*$ shows a similar increasing trend to U_j^* and U_f^* with the decrease in R and the increase in H/W. The horizontal velocity comes from the eccentric impact of the fallen droplet. As U_j^* and U_f^* increase with the decrease in R and the increase in H/W, the reaction force of the eccentric impact is enhanced, which results in the increase in the horizontal velocity. The maximum value of $U_{r,x}^*$ is 0.47 when $R = 0.66$ mm and $H/W = 2.0$, and the minimum value is 0.20 when $R = 1.25$ mm and $H/W = 1.0$. Unlike $U_{r,x}^*$, $U_{r,y}^*$ increases with the decrease in R but fluctuates with the change in H/W. The horizontal velocity can be up to 2.3 times the vertical jumping velocity of the jumping droplet on the plane, which indicates effective enhancement of the horizontal mobility of the coalesced jumping droplet.

In the experiment, a horizontal offset Δx of the centroid of the fallen droplet relative to the ridge is observed when the fallen droplet makes contact with the ridge. For comparison, the ratio of the horizontal offset to the initial droplet radius $\Delta x/R$ is used to represent the relative horizontal offset of the centroid of the fallen droplet. Figure 5(a) shows the variation in the relative horizontal offset $\Delta x/R$ with the initial droplet radius R and the height-to-width ratio H/W. $\Delta x/R$ decreases as R increases but changes little for different H/W. Because of the horizontal offset of the droplet centroid, the fallen droplet impacts the asymmetric ridge eccentrically, leading to the directional rebound of the droplet. The variation in the ratio of the nondimensional horizontal velocity $U_{r,x}^*$ to the nondimensional fallen velocity U_f^* with R and H/W is shown in Fig. 5(b). $U_{r,x}^*/U_f^*$ varies little with R but increases with H/W. When H/W increases from 1.0 to 2.0, $U_{r,x}^*/U_f^*$ increases from about 0.55 to about 0.75, which indicates that a considerable proportion of the fallen velocity is converted into the horizontal velocity, and the proportion can be regulated by tuning the H/W of the asymmetric ridge.

B. Dynamic analysis

To reveal the mechanism underlying the enhanced horizontal mobility, we analyze the droplet dynamics of the coalescence-induced jumping stage and the rebound stage using numerical simulation. The numerical results for the coalescence-induced jumping stage are shown in Fig. 6. Figure 6(a) shows the variation in the vertical component of the nondimensional jumping velocity $U_{j,y}{}^*$ and the nondimensional vertical force $F_{j,y}{}^*$ with time, Fig. 6(b) shows the variation in the horizontal component of the nondimensional jumping velocity $U_{j,x}{}^*$ and the nondimensional horizontal force $F_{j,x}{}^*$ with time, and Fig. 6(c) shows the pressure and velocity fields in the X–Y plane inside the

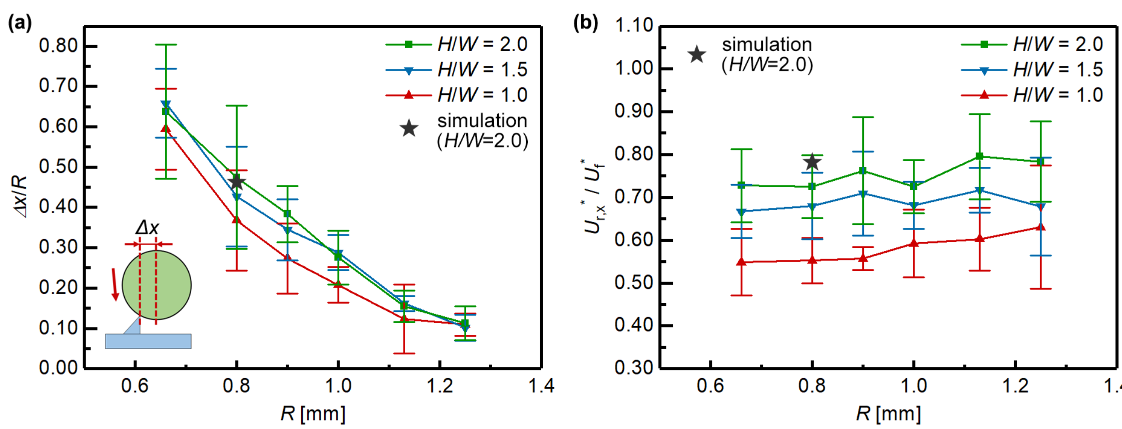

FIG. 5. (a) The relative horizontal offset $\Delta x/R$ of the centroid of the fallen droplet to the ridge. The $\Delta x/R$ in the simulation is the same value of the experimental result at $R = 0.8$ mm. (b) The ratio of the nondimensional horizontal velocity and the nondimensional fallen velocity $U_{r,x}{}^*/U_f{}^*$. The star denotes the numerical results.

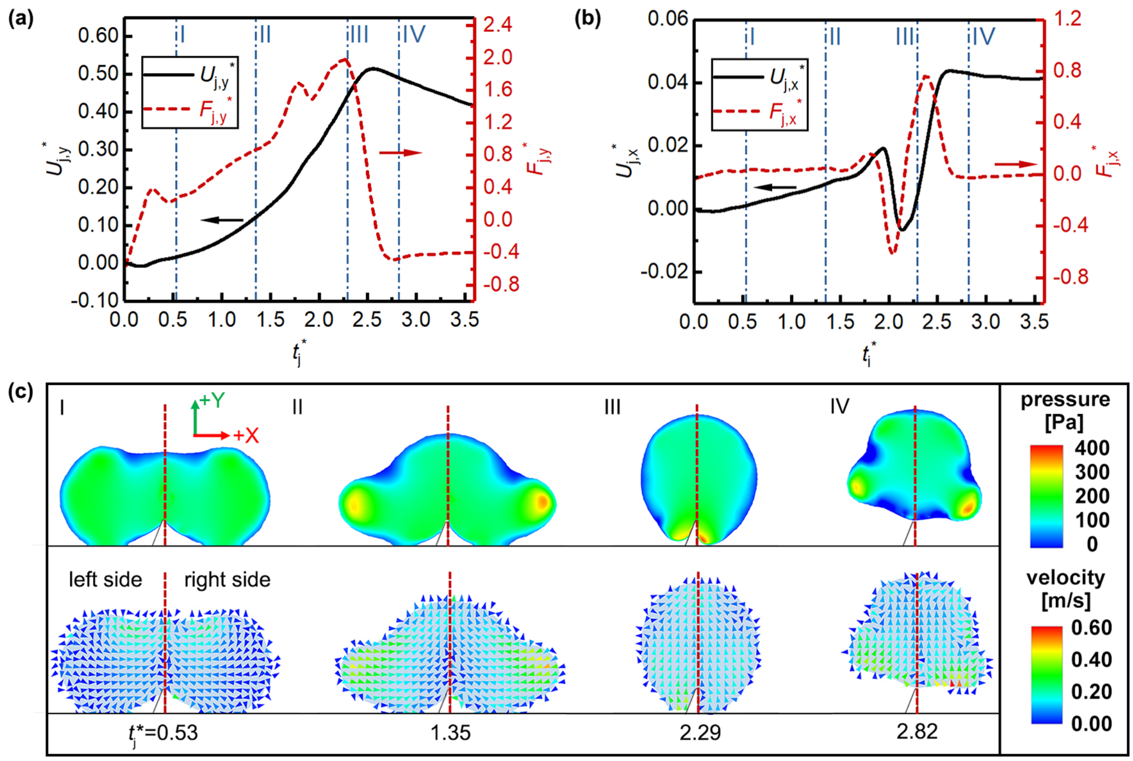

FIG. 6. Droplet dynamics of the coalescence-induced jumping stage. (a) Variation in the vertical component of the nondimensional jumping velocity $U_{j,y}{}^*$ and the nondimensional vertical force $F_{j,y}{}^*$ with time. (b) Variation in the horizontal component of the nondimensional jumping velocity $U_{j,x}{}^*$ and the nondimensional horizontal force $F_{j,x}{}^*$ with time. (c) Pressure and velocity fields in the X–Y plane inside the droplet at the corresponding moments of I–IV.

droplet at the corresponding moments of I–IV in Figs. 6(a) and 6(b). At the onset of coalescence, the liquid bridge forms where the two droplets make contact. The fluid flow is driven toward the liquid bridge by the capillary pressure, which causes the liquid bridge to expand symmetrically. At $t_j^* = 0.53$ [Fig. 6(c-I)], the liquid bridge makes contact with the top of the ridge, and a high-pressure region forms near the top of the ridge. The interaction between the liquid bridge and the ridge generates an upward force, causing the droplet to move upward. Subsequently, at $t_j^* = 1.35$ [Fig. 6(c-II)], high-pressure regions form on both sides of the coalesced droplet, which causes the fluid on both sides of the droplet to contract toward the middle. The internal flow is redirected to the +Y direction under the effect of the ridge,[32] which leads to an increase in the upward velocity $U_{j,y}{}^*$. From $t_j^* = 1.35$ to $t_j^* = 2.29$, $U_{j,x}{}^*$ first increases slowly and then decreases rapidly to a negative value. This is because the asymmetry of the ridge makes the fluid on the left side make contact with the left slope of the ridge (left side of the red dotted line) and a reaction force is generated, whereas the fluid on the right side has not made contact with the ridge and no reaction force is generated on the right side. Under the effect of the reaction force of the left slope, $F_{j,x}{}^*$ increases rapidly along the −X direction. The velocity vectors of the internal flow on the left side are redirected by the left slope, whereas most of the velocity vectors on the right side are still toward the ridge, which makes $U_{j,x}{}^*$ turn to the −X direction. At $t_j^* = 2.29$ [Fig. 6(c-III)], the coalesced droplet wraps around the ridge and a local high-pressure zone forms around the ridge. Because of the contact delay between the liquid on the right side and the ridge, most of the local high-pressure zone is located on the right side of the ridge. Subsequently, $F_{j,x}{}^*$ turns back to the +X direction, which results in a change in the direction of $U_{j,x}{}^*$. At $t_j^* = 2.82$ [Fig. 6(c-IV)], obvious asymmetry is shown in the internal pressure and velocity fields, and the coalesced droplet detaches from the surface with $U_{j,y}{}^* = 0.49$ and $U_{j,x}{}^* = 0.04$. The horizontal component of the jumping velocity makes the droplet have a horizontal offset of the centroid and eventually impact the ridge eccentrically. The nondimensional jumping velocity from the numerical simulation is 0.49, with the corresponding experimental results of 0.60.

Figure 7 shows the numerical results for the rebound stage. Figure 7(a) shows the variation in the nondimensional vertical rebound velocity $U_{r,y}^*$ and the nondimensional vertical reaction force $F_{r,y}{}^*$ with time, Fig. 7(b) shows the variation in the nondimensional horizontal velocity $U_{r,x}^*$ and the nondimensional horizontal reaction force $F_{r,x}{}^*$ with time, and Fig. 7(c) shows the pressure and velocity fields in the X–Y plane inside the droplet at the corresponding moments of I–IV in Figs. 7(a) and 7(b). At $t_r^* = 0.35$ [Fig. 7(c-I)], the fallen droplet has made contact with the ridge but not yet the plane. Owing to the horizontal offset Δx between the centroid of the fallen droplet and the ridge, the droplet volume on the right side of the ridge is greater. The high-pressure region at the bottom of the droplet shows clearly that the pressure on the left side of the ridge is greater than that on the right side, generating both vertical and horizontal reaction

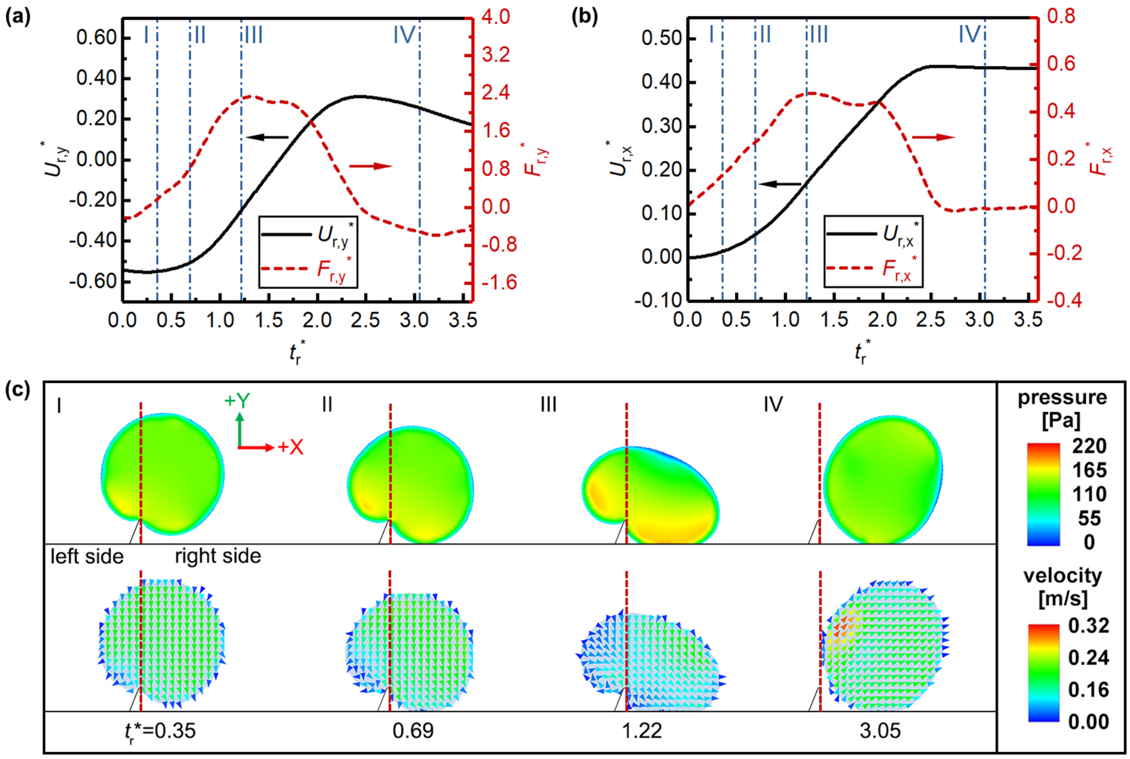

FIG. 7. Droplet dynamics of the rebound stage. (a) Variation in the vertical component of the nondimensional jumping velocity $U_{j,y}{}^*$ and the nondimensional vertical force $F_{j,y}{}^*$ with time. (b) Variation in the horizontal component of the nondimensional jumping velocity $U_{j,x}{}^*$ and the nondimensional horizontal force $F_{j,x}{}^*$ with time. (c) Pressure and velocity fields in the X–Y plane inside the droplet at the corresponding moments of I–IV.

force. The internal flow on the left side is slowed down by the Laplace pressure generated by the asymmetric deformation of the droplet. At $t_r^* = 0.69$ [Fig. 7(c-II)], the droplet continues to move downward, and the right side of the droplet makes contact with the plane, resulting in an expansion of the high-pressure region on both sides of the ridge. Because of the squeezing of the ridge, the left side of the droplet is lifted, and the deformation of the left side generates horizontal force. Consequently, both $F_{r,y}{}^*$ and $F_{r,x}{}^*$ increase. At $t_r^* = 1.22$ [Fig. 7(c-III)], the droplet spreads on the surface, and the Laplace pressure on the left side drives the internal fluid to accelerate toward +X direction. Both the vertical and horizontal reaction force reach the maximum ($F_{r,y}{}^* \approx 2.4$, $F_{r,x}{}^* \approx 0.5$), which leads to a rapid change in $U_{r,y}^*$ and $U_{r,x}^*$. At $t_r^* = 3.05$ [Fig. 7(c-IV)], the droplet rebounds from the ridge. Compared to moment I, the velocity vector of the droplet deviates from the perpendicular direction because of the effects of the asymmetric ridge on the droplet rebound process, resulting in the horizontal velocity. The numerical results ($U_{r,x}^* = 0.43$, $U_{r,x}^*/U_f^* = 0.78$) are in good agreement with the experimental results ($U_{r,x}^* \approx 0.41$, $U_{r,x}^*/U_f^* \approx 0.73$), which verifies the reliability of the simulation.

C. Energy analysis

To further illustrate the mechanism underlying the enhanced horizontal mobility, we analyze energy conversion in the coalescence-induced jumping stage and the rebound stage using numerical simulation. The energy conversion of the droplet in motion is governed by the released surface energy, the jumping kinetic energy, viscous dissipation, and gravitational potential energy. Note that the effect of gravitational potential energy can be ignored for two reasons. First, $Bo < 0.10$ ($Bo = \rho_l R^2/\gamma_{lv}$) for droplets in the experiment and simulation. Second, the change in the height of the droplet centroid in the coalescence-induced jumping stage and the rebound stage is negligible. The released surface energy $\Delta E_s(t)$ is calculated as follows:

$$\Delta E_s(t) = \gamma_{lv}(A_{lv}(t) - 4\pi R_e^2), \tag{10}$$

where t is the time, γ_{lv} is the surface tension, $A_{lv}(t)$ is the surface area of the droplet at moment t, and R_e is the equivalent radius of the coalesced droplet.[33]

The jumping kinetic energy $E_k(t)$ is calculated as follows:

$$E_k(t) = \frac{mV^2(t)}{2}, \tag{11}$$

where V is the jumping velocity of the droplet and m is the mass of the coalesced droplet.

The viscous dissipation $E_{vis}(t)$ is calculated as follows:

$$E_{vis}(t) = \int_0^t \int_\Omega \phi \, d\Omega \, dt, \tag{12}$$

$$\phi = \mu\left[2\left(\frac{\partial u}{\partial x}\right)^2 + 2\left(\frac{\partial v}{\partial y}\right)^2 + 2\left(\frac{\partial w}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial z}\right)^2\right], \tag{13}$$

where μ is the viscosity of water and w is the velocity components of the internal flow in the Z direction. The released surface energy, jumping kinetic energy, and viscous dissipation are nondimensionalized as follows:

$$E_k^*(t) = \frac{E_k(t)}{E(0)}, \tag{14}$$

$$\Delta E_s^*(t) = \frac{\Delta E_s(t)}{E(0)}, \tag{15}$$

$$E_{vis}^*(t) = \frac{E_{vis}(t)}{E(0)}, \tag{16}$$

where $E(0)$ is the total energy of the droplets at the initial moment. For the coalescence-induced jumping stage, $E(0) = \Delta E_s(0)$. For the rebound stage, $E(0) = E_k(0)$. The energy conversion efficiency $\eta_{f,x}$ of the fallen kinetic energy to the horizontal kinetic energy and the energy conversion efficiency $\eta_{s,x}$ of the released surface energy to the horizontal kinetic energy are calculated as follows:

$$\eta_{f,x} = \frac{mV_{r,x}^2/2}{mV_f^2/2}, \tag{17}$$

$$\eta_{s,x} = \frac{mV_{r,x}^2/2}{\gamma_{lv}(A_{lv}(0) - 4\pi R_e^2)}, \tag{18}$$

where $V_{r,x}$ is the horizontal velocity, V_f is the velocity when the fallen jumping droplet is about to contact the top of the ridge, and 0 represents the start of the coalescence.

Figure 8(a) shows the variation in energy with time in the coalescence-induced jumping stage. The change in the nondimensional released surface energy ΔE_s^* exhibits a clear decreasing trend. The reduced ΔE_s^* is converted into the jumping kinetic energy E_k^* and the viscous dissipation E_{vis}^*. At $t_j^* = 1.25$–1.85, ΔE_s^* decreases rapidly and E_k^* increases simultaneously because of the contraction of the coalescing droplet.[16] After the coalesced droplet detaches from the surface ($t_j^* = 2.82$), the energy conversion rate η of the released surface energy into the jumping kinetic energy is about 25%, which increases by approximately 400% compared to that of the droplet jumping on the plane ($\eta \approx 5\%$). Figure 8(b) shows the variation in energy with time in the rebound stage. At $t_r^* = 0$–1.50, the fallen droplet moves downward and spreads on the surface. The change in the internal flow velocity leads to a continued increase in the viscous dissipation E_{vis}^* and deformation of the droplet. As a result of the deformation of the droplet, the jumping kinetic energy is converted into surface energy, which results in a decrease in E_k^* and an increase in ΔE_s^*. After reaching the maximum spreading state ($t_r^* = 1.50$–3.05), the fallen droplet retracts, with the released surface energy converting back to the jumping kinetic energy. At the moment of detachment ($t_r^* = 3.05$), ΔE_s^* is close to 0, which indicates that the surface energy is nearly unchanged before and after the rebound, which is consistent with the characteristics of the small deformation process. For jumping kinetic energy, $E_k^* \approx 0.87$ at the moment of detachment, which indicates that about 87% of the initial jumping kinetic energy is reserved, with the rest dissipated by the viscous effect.

Figure 9 shows the variation in the ratio $E_k(t_d)/E_k(0)$ of the jumping kinetic energy at the moment of detachment from the ridge and at the initial time with R and H/W in the rebound stage, where t_d is the time when the fallen droplet detaches from the ridge. $E_k(t_d)/E_k(0)$ ranges from 0.74 to 0.86, which shows that $E_k(t_d)/E_k(0)$ has no obvious correlation with R and H/W. The experimental results match well with the numerical results [$E_k(t_d)/E_k(0) = 0.87$], which indicates that the loss in jumping kinetic energy is around 20% during the rebound stage.

Figure 10 shows the variation in the energy conversion efficiency with R and H/W. As shown in Fig. 10(a), the energy conversion efficiency $\eta_{f,x}$ of the fallen kinetic energy to the horizontal kinetic energy increases with the increase of H/W but has no obvious change with R. As for the energy conversion efficiency $\eta_{s,x}$ of the released surface energy to the horizontal kinetic energy [Fig. 10(b)], $\eta_{s,x}$ increases with the increase of H/W and decreases with the increase of R. This is

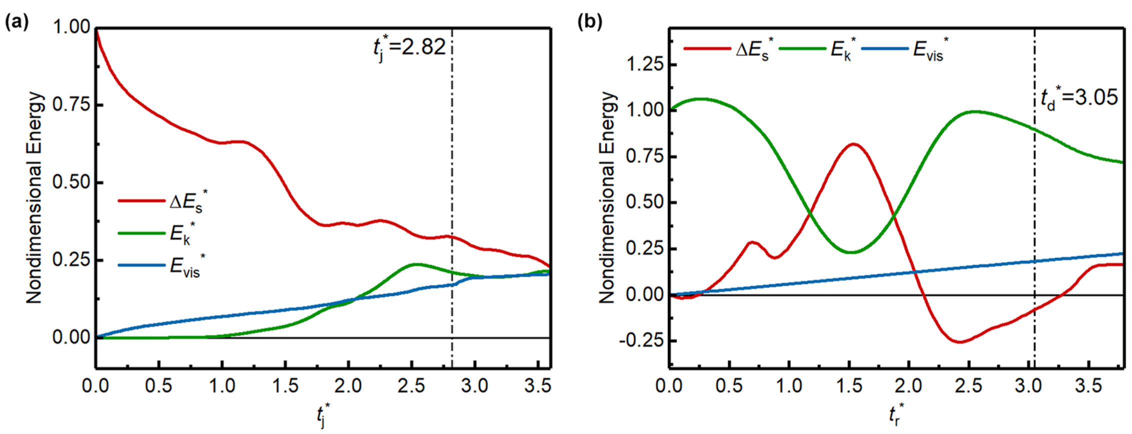

FIG. 8. Variation in nondimensional energy with time, where the dotted line denotes the moment of detachment: (a) coalescence-induced jumping stage and (b) rebound stage.

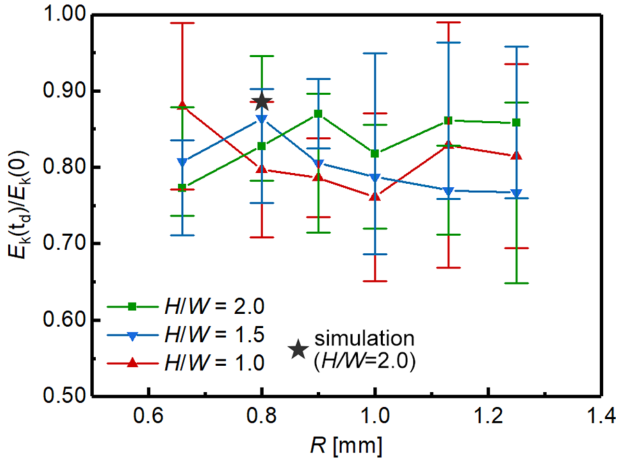

FIG. 9. Ratio of jumping kinetic energy at the moment of detachment and at the initial time with R and H/W in the rebound stage, where the star denotes the numerical results.

caused by the variation in the jumping kinetic energy converted from the released surface energy. According to Ref. 33, the jumping kinetic energy converted from the released surface energy increases with the increase of H and decreases with the increase of R. As shown in Fig. 4(b), the fallen velocity is basically the same as the jumping velocity, which indicates that the fallen kinetic energy is also basically the same as the jumping kinetic energy. Considering the variation in $\eta_{f,x}$ shown in Fig. 10(a), the variation in $\eta_{s,x}$ is similar to the trend of the jumping kinetic energy converted from the released surface energy.

D. Effect of the initial droplet radius

The coalescence of a droplet with unequal initial radii is ubiquitous in practical applications. Therefore, the motion of droplets with different initial radii is investigated experimentally. The radii of droplets on the perpendicular and inclined sides of the ridge are noted R_p and R_i, respectively, and the initial droplet radius ratio is noted R_p/R_i. R_p is fixed at 0.89 mm, and R_i ranges from 0.66 to 1.25 mm, with R_p/R_i ranging from 0.75 to 1.41. Figure 11(a) (Multimedia view) shows the entire motion of the coalesced jumping droplet with different initial droplet radii on superhydrophobic surfaces with an asymmetric ridge ($W = 0.2$ mm; $H/W = 1.5$; and $R_p/R_i = 0.75$, 0.91, 1.16, 1.43). The schematic of the position of the initial droplets and the coordinate system is shown in Fig. 11(b). When the initial droplet radius ratio changes, the motion of the coalesced droplet still consists of three stages: coalescence-induced jumping, detachment, and rebound. Unlike the coalescence of two equal-size droplets, the coalescence of two unequal-size droplets exhibits significant asymmetry in that the droplet with the smaller radius contracts faster toward the middle than the larger one.[66] Therefore, the centroid of the coalesced droplet is closer to the side of the larger droplet when it detaches from the ridge, and the fallen droplet rebounds toward the side of the larger droplet. Figure 11(c) shows the variation in the horizontal velocity $U^*_{r,x}$ with R_p/R_i. The coalesced jumping droplet is transported to the inclined side of the ridge (the +X direction) when $R_p/R_i \leq 1.0$ and the perpendicular side of the ridge (the −X direction) when $R_p/R_i > 1.0$, which shows a directionality related to R_p/R_i. The nondimensional horizontal velocity $U^*_{r,x}$ increases as the difference in the initial droplet radius decreases for both $R_p/R_i \leq 1.0$ and $R_p/R_i > 1.0$. For the same two unequal-size droplets, the horizontal velocity ($U^*_{r,x} = 0.20$–0.44) when the large droplet is on the inclined side of the ridge is greater than that when the large droplet is on the perpendicular side ($U^*_{r,x} = 0.10$–0.38), which indicates that the inclined side allows higher horizontal velocity of the coalesced droplet. As for the energy conversion efficiency shown in Figs. 11(a) and 11(b), $\eta_{f,x}$ varies little when $R_p/R_i \leq 1.0$ and decreases with R_p/R_i when $R_p/R_i > 1.0$ and $\eta_{s,x}$ shows similar trend with $U^*_{r,x}$. Nevertheless, the minimum horizontal velocity of both cases is comparable to the jumping velocity ($U^*_j \approx 0.20$) of two equal-size droplets coalescing on the superhydrophobic plane, which demonstrates that the ridge can enhance the horizontal mobility of the coalesced jumping droplet with various initial droplet radius ratios.

IV. CONCLUSION

Here, we report enhanced horizontal mobility of a coalesced jumping droplet on superhydrophobic surfaces with an asymmetric

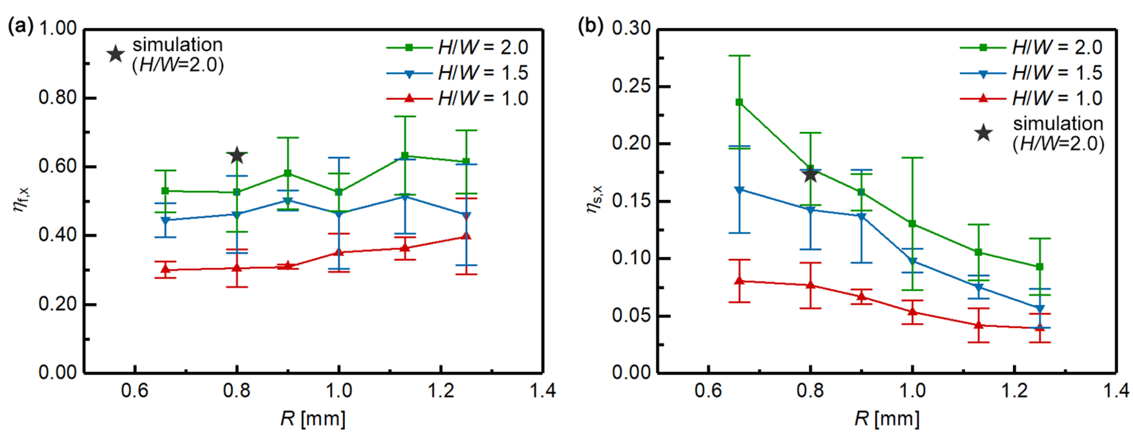

FIG. 10. Variation in the energy conversion efficiency with R and H/W. (a) The energy conversion efficiency $\eta_{f,x}$ of the fallen kinetic energy to the horizontal kinetic energy. (b) The energy conversion efficiency $\eta_{s,x}$ of the released surface energy to the horizontal kinetic energy. The star denotes the numerical results.

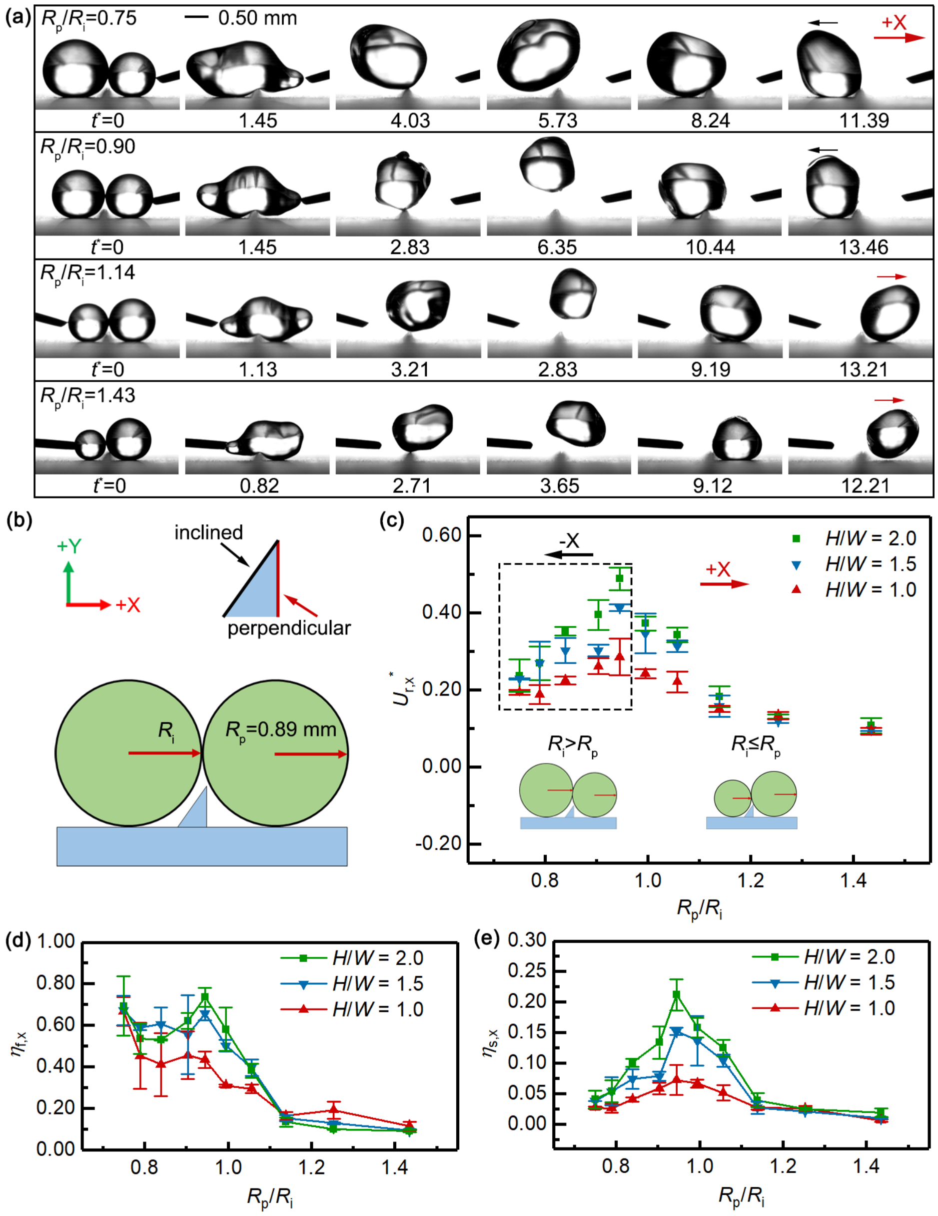

FIG. 11. (a) Motion of a coalesced jumping droplet with different initial droplet radii on superhydrophobic surfaces with an asymmetric ridge ($W = 0.2$ mm; $H/W = 1.5$; and $R_p/R_i = 0.75$, 0.90, 1.14, 1.43). (b) Schematic of the position of the initial droplets and the coordinate system. (c) Variation in the nondimensional horizontal velocity $U_{r,x}^*$ with the initial droplet radius ratio R_p/R_i. (d) Variation in the energy conversion efficiency $\eta_{f,x}$ of the fallen kinetic energy to the horizontal kinetic energy with R_p/R_i. (e) The energy conversion efficiency $\eta_{s,x}$ of the released surface energy to the horizontal kinetic energy with R_p/R_i. Multimedia views; https://doi.org/10.1063/5.0121402.2; https://doi.org/10.1063/5.0121402.3; https://doi.org/10.1063/5.0121402.4; https://doi.org/10.1063/5.0121402.5

ridge and the results of systematic investigations of the mechanism underlying the enhancement. Experimental results show that the coupling effects of the asymmetric ridge on the stages of coalescence-induced jumping and the fallen droplet contacting the surface lead to the enhanced horizontal mobility. The maximum nondimensional horizontal velocity $U_{r,x}^*$ can reach 0.47, and the ratio of the horizontal velocity to the fallen velocity can be regulated by adjusting the height-to-width ratio of the asymmetric ridge. Through numerical simulation, we reveal that the coalesced jumping droplet achieves enhanced jumping velocity as a result of a redirection of the internal flow in the coalescence-induced stage and achieves horizontal velocity as a result of an asymmetric high-pressure region in the horizontal direction under the effect of the ridge in the rebound stage. The coalesced droplet gets the kinetic energy from the released surface energy during coalescence-induced jumping process and then gets the horizontal kinetic energy during the rebound process. Furthermore, we demonstrate experimentally that a coalesced droplet with different initial droplet radii can still move with considerable horizontal velocity, and the coalesced droplet rebounds to the side of the larger droplet. This work demonstrates that ridge-like structure can also enhance the horizontal mobility of a coalesced jumping droplet when the droplet falls back to the surface. Those findings might be helpful for promoting droplet jumping and improving the performance of related applications.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Sihang Gao: Conceptualization (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Validation (lead); Writing – original draft (lead). **Zhifeng Hu:** Methodology (equal); Validation (equal); Writing – review & editing (equal). **Xiaomin Wu:** Conceptualization (supporting); Funding acquisition (lead); Project administration (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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